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# Zhengchun Liu

Applied Scientist, PhD

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Zhengchun has 10+ years of experience in Research and Development, specializing in Data Infrastructure, Machine Learning, and High-Performance Computing systems. Since 2022, he has been serving as a Senior Applied Scientist at AWS, where he Leads the Technical development of ML capabilities in Redshift, focusing on enabling autonomous capabilities for data warehouse. Before this, he was a Tenured Computer Scientist at Argonne National Laboratory from 2016 to 2024 where he led or co-led 10+ NSF/DOE funded research projects, secured over $12M in funding from government agencies. His expertise spans technical leadership, innovative problem-solving, and advancing cutting-edge technologies in his field.

## Research & Engineering Experience

**Amazon Web Service (AWS), AI Labs** **Aug. 2022 — present**
*Senior Applied Scientist*
- Tech Lead (Initiate, Design & Review) of Machine Learning model for AI-driven Autonomous data warehouse.
- The ML Models developed support billions of decisions for Scaling, Scheduling and Optimization every day.
- Drive predictive analytics of customer behavior, and launch data-driven initiatives for customer pain points.
- Lead scientist developing the Agentic AI system that powers analytic SQL generation in Amazon Ads Agent.

**Argonne National Laboratory, Data Science and Learning division** **Aug. 2019 — Aug. 2024**
*Sabbatical leave (Aug. 2022), Tenured Computer Scientist (May 2022), Assistant Computer Scientist (Aug. 2019)*
- Serve as a Principal Investigator (PI) or co-PI of 10+ NSF/DOE research projects totaling more than $12M of funding.
- Lead a team conducting research on Artificial Intelligence for Science (AI4S), and HPC/Systems for AI4S applications.
- Lead author of over 20 Publications in prestigious venues on Data Infrastructure, HPC and AI4S.

**University of Chicago, Computation Institute** **Mar. 2018 — Aug. 2019**
*Research Scientist*
- Senior Personnel or co-PI of research projects on the architecture of Autonomous science ecosystems.
- Drive the research of performance modeling and characterization of scientific applications on supercomputers.

**Argonne National Laboratory, Mathematics and Computer Science division** **Sep. 2016 — Mar. 2018**
*Postdoctoral Researcher*
- Led research on High Performance File Transfer over Wide-Area Science Network.
- Key Personnel of a project on Modeling, Simulation and Optimization of scientific workflows over distributed infra.

***Founder & Operator**; Embedded System Technology Consulting & Outsourcing, China* *Sep. 2010 — Nov. 2013*
- 10+ Consulting and/or Outsourcing contracts, three prototypes went into massive production.
- Board Support Package development for $\mu$C/OS-II real-time operating system.
- Designed and Led the implementation of an event streaming service for Vehicle Event Data Recorder (connected IoT devices).

## Selected Awards, Grants & Honors

- `Best Paper` award $\times$**5**; XLOOP@SC'21, FGCS'21, MLN'19, MLN'18, and TRIDENTCOM'18 — 2021, 2019, 2018
- `Impact Argonne/Pacesetter` award $\times$**3**; for notable achievement in `Innovation` — Nov. 2019, May 2020, July 2022
- `Top Winner` of the Technology Challenge at the ACM/IEEE Supercomputing Conference (SC'19) — Nov. 2019
- `Extraordinary Doctorate Award` by Autonomous University of Barcelona — Jun. 2018
- The 1st Place in the `10th Marathon of Parallel Programming Contest` — Oct. 2015
- The 1st Place in the `5th Spanish Parallel Programming Contest` — Sep. 2015

## Education

- **PhD., Computer Science**, Autonomous University of Barcelona, Spain — Jul. 2016
  - `With Honors(Cum Laude); Extraordinary Doctorate Award(1 out of 5 thesis).`
- **MSc., Guidance, Navigation and Control**, Northwestern Polytechnical University, China — Apr. 2013
- **BSc., Aircraft Manufacturing Engineering**, Northwestern Polytechnical University, China — Jun. 2010
  - `National Scholarship` (award to top 0.2% of outstanding undergraduates) $\times$2, 2008 and 2009.
  - `Champion` of the Chinese Robot Competition (RoboCup 2008), dancing session. Dec. 2008

## Activities

- `Editorship`: Journal of Future Generation Computer Systems (FGCS), 2020-2023, Impact Factor: 7.2.
- `Workshop Co-Chair`: SRMPDS 2017, 2018, 2019, 2020; AI-Science'19.
- `Technical Program Committee`: ICDS'19; DAAC'17-19; DLS'19; ICDCS'20; SC'20; HPCC'20; HiPC'22; SC'23; IPDPS'23,24.